
OpenAI Model Craft Challenge: Parameter Golf

CSCI.GA-2565 Machine Learning

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Abstract

We tackle the OpenAI “Parameter Golf” challenge, an extreme $\mathcal{L}(N)$ optimization problem that minimizes language model bits-per-byte within a strict 16 MB self-contained artifact budget. To maximize parameter efficiency, we conduct a rigorous architecture search evaluating macro-topological compression techniques, including depth recurrence, residual subspace projections, and learned inter-layer connectivity. We also introduce **k-XSA**, a novel rank- k exclusive self-attention mechanism featuring a quantization-aware energy-cap regularizer. Our findings show that macroscopic parameter sharing fails due to the necessity of depth-specific block specialization, whereas topological simplifications such as our “NoSkip” backbone yield a strict Pareto improvement. Furthermore, k-XSA consistently outperforms standard XSA and exhibits robustness to aggressive low-precision quantization, mapping the effective boundary between architectural capacity and quantization limits at the extreme-compression frontier.

1 Introduction

The dominant paradigm in LLM research treats parameter count as a free variable, minimizing loss as a function of compute $\mathcal{L}(C)$ Kaplan et al. [2020], Hoffmann et al. [2022] or dataset size $\mathcal{L}(D)$. This challenge focuses $\mathcal{L}(N)$ problem: *minimize held-out loss when N is fixed by a hard storage constraint*. The OpenAI “Parameter Golf” challenge OpenAI [2026] requires the submission to be a self-contained artifact of at most **16 MB** (code plus weights), scored by **BPB** on the FineWeb Penedo et al. [2024] validation set, with training capped at **10 minutes on $8\times\text{H100 GPUs}$** .

The baseline model is a 9-layer, 512-dim transformer with 1024-token BPE, GQA, and the Muon optimizer which scores 1.2244 BPB at 15.86 MB. The leaderboard SOTA (1.1194 BPB, March 2026) uses LeakyReLU², test-time training Sun et al. [2024], and 11 layers; other top entries use INT6 QAT, Exclusive Self Attention (XSA) Zhai [2026], and EMA weight averaging. Operating at this extreme demands compression-first design: every embedding dimension, vocabulary size, and quantization scheme must be evaluated through the lens of performance-gain per stored byte.

We explore this space with an asymmetric encoder-decoder using GQA Ainslie et al. [2023], RoPE Su et al. [2022], RMSNorm, and ReLU². We ablate three compression axes: (i) **recurrency**: one shared block reused across nine iterations with per-step bias; (ii) **subspace projection**: residual stream routed through a $D \rightarrow k \rightarrow D$ bottleneck; and (iii) **aggregate connectivity**: learned per-channel mixtures over all prior hidden states. We also propose **K-XSA**, extending Zhai’s XSA Zhai [2026] to a learned rank- k orthogonal subspace with a per-head learnable α and a quantization-aware energy-cap regularizer. Key findings: (1) a non-recurrent *NoSkip* baseline with nine distinct blocks wins overall; (2) depth-specific specialization per-block attention usage ramps $33\times$ with depth cannot be recovered by weight tying; (3) K-XSA improves over XSA at every tested rank, with learnable α as the dominant factor; (4) int6-g64 quantization and SP4096 vocabulary are the sweet spots near the 16 MB boundary.

2 Related Work

Our work intersects four literatures.

2.1 Compression-first language modeling and the $\mathcal{L}(N)$ regime

Empirical scaling laws Kaplan et al. [2020], Hoffmann et al. [2022] treat N as a free hyperparameter optimized over compute or data. The dual $\mathcal{L}(N)$ problem fixing N via a hard storage constraint has received less systematic attention. Compact LM efforts such as MobileLLM Liu et al. [2024], Phi Gunasekar et al. [2023], and TinyStories Eldan and Li [2023] target budgets orders of magnitude above 16 MB and rely on post-hoc compression rather than compression-first training. Parameter Golf OpenAI [2026] is the $\mathcal{L}(N)$ analogue of the NanoGPT speedrun Karpathy [2024]; to our knowledge, no prior work characterizes where BPB headroom localizes at this scale.

2.2 Quantization-aware design

Post-training quantization Frantar et al. [2022] and quantization-aware training Jacob et al. [2018] are well-established, and sub-2-bit weights are viable at scale Wang et al. [2023], Ma et al. [2024]. Group-wise quantization Frantar et al. [2022] reduces low-bit precision loss but has not, to our knowledge, been characterized as a function of tokenizer choice at the 16 MB scale. Our work treats quantization as a primary design axis: the energy cap of Section 3.4.4 acts as a quantization-aware regularizer, connecting loosely to flat-minima and sharpness penalty literatures, though this property emerged empirically rather than by design.

2.3 Compact attention operators

Multi-Query and Grouped-Query Attention Shazeer [2019], Ainslie et al. [2023] and RoPE Su et al. [2022] reduce attention parameter cost; our baseline uses both. Closer to our third study, Apple’s Exclusive Self-Attention Zhai [2026] modifies the attention *output* by stripping its rank-1 component along v_i . Our k-XSA (Section 3.4) is a strict generalization that lifts this to a rank- k projection in a learned subspace and adds a per-direction soft-threshold energy cap; we are not aware of prior work combining these two ingredients in the same operator.

3 Method

3.1 Setup, notation, and accounting

Backbone. All three studies share the asymmetric encoder–decoder transformer of Figure 1: tied token embedding, input RMSNorm, n_{enc} encoder blocks whose post-block residuals are saved as U-Net skip states, n_{dec} decoder blocks consuming those skips in reverse, output RMSNorm, and tied LM head. Each block uses GQA+RoPE attention and a squared-ReLU MLP of expansion factor 2. The per-block forward is

$$\tilde{x} = \mu_0 x + \mu_1 x_0, \quad x \leftarrow \tilde{x} + \sigma_\ell^{\text{attn}} \text{Attn}(\tilde{x}), \quad x \leftarrow x + \sigma_\ell^{\text{mlp}} \text{MLP}(\text{ReLU}(x)^2), \quad (1)$$

where x_0 is the post-RMSNorm token embedding, $\mu \in \mathbb{R}^2$ are scalar residual-mixing weights, and $\sigma_\ell^{\text{attn}}, \sigma_\ell^{\text{mlp}} \in \mathbb{R}$ are zero-initialized per-block scales. The U-Net branch contributes a channel-wise skip $\text{skip}_d \odot h^{(n_{\text{enc}}-d)}$ with $\text{skip}_d \in \mathbb{R}^D$ also zero-initialized. The ablation and vocab–quant sweep use $(n_{\text{enc}}, n_{\text{dec}}, D) = (4, 5, 512)$; k-XSA uses $(5, 6, 512)$ to match recent leaderboard configurations.

3.2 Architectural variants

3.2.1 Phase 1: alternatives to the baseline topology

Weight-tied recurrence (Recurrent). Replace $L = n_{\text{enc}} + n_{\text{dec}}$ distinct blocks with one shared block called L times, plus per-iteration biases $\mathbf{e} \in \mathbb{R}^{L \times D}$:

$$h^{(\ell+1)} = \text{Block}(h^{(\ell)} + \mathbf{e}_\ell), \quad \ell = 0, \dots, L-1. \quad (2)$$

Operationalizes “depth is a compute axis, not a specialization axis.”

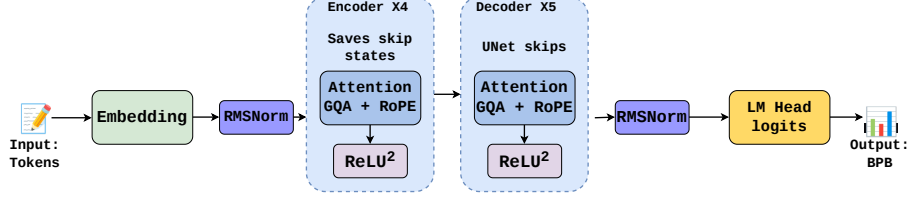


Figure 1: Baseline asymmetric encoder-decoder transformer shared by all studies: tied embedding, $n_{\text{enc}}=4$ encoder blocks feeding U-Net cross-skips into $n_{\text{dec}}=5$ decoder blocks, GQA+RoPE attention and squared-ReLU MLP per block (Equation (1)).

Low-rank residual subspace (Subspace). Each block runs at smaller dimension k :

$$h^{(\ell+1)} = h^{(\ell)} + W_{\uparrow}^{(\ell)} [\text{Attn}_k + \text{MLP}_k] (W_{\downarrow}^{(\ell)} h^{(\ell)}), \quad k \in \{D/2, D/4\}. \quad (3)$$

Operationalizes “each block specializes in a k -dimensional subspace.”

Learned aggregate connectivity (Aggregate). Block ℓ consumes a per-channel weighted sum of all prior residuals:

$$\text{inp}^{(\ell)} = \sum_{i=0}^{\ell} A_{\ell,i} \odot h^{(i)}, \quad A \in \mathbb{R}^{L \times (L+1) \times D}, \quad (4)$$

which contains residual stacks, DenseNets, and U-Nets as special cases. The per-block μ in Equation (1) is disabled since A subsumes its role.

3.2.2 Phase 2: additive refinements to the baseline

Per-channel embedding gate (EmbedGate). Replace scalar μ_1 in Equation (1) with a per-channel gate $g_{\ell} \in \mathbb{R}^D$: $x \leftarrow x + g_{\ell} \odot x_0$, zero-initialized for bit-exact baseline at step zero. Tests whether Aggregate’s converged “every block reads h_0 ” pattern, expressed per-channel, yields a gain.

Removal of U-Net skip weights (NoSkip). Remove the four U-Net skip_weights: they converge to small or negative magnitudes ($\{0.494, 0.317, 0.442, -0.019\}$ in baseline) a confounder signature, not a contributor.

Block-0 subspace bottleneck (Block0Subspace). Apply Equation (3) ($k = D/2$) only at $\ell = 0$, since $\sigma_0^{\text{attn}} = 0.144$ is $33\times$ smaller than $\sigma_8^{\text{attn}} = 4.563$ in baseline.

We also evaluate two combinations (EmbedGate+NoSkip, all-three).

3.3 Vocabulary and quantization-scheme sweep

The byte budget is controlled jointly by tokenizer vocabulary (determining the embedding table and the bpb floor via) and post-training quantization (determining the bytes-per-parameter cost). These axes interact multiplicatively at the cap.

Vocabulary axis. We train four model variants from scratch under the baseline architecture, varying only the SentencePiece BPE vocabulary $|V| \in \{1024, 2048, 4096, 8192\}$. The non-embedding parameter count is held constant at 16.53 M; embedding parameters scale from 0.52 M ($|V|=1024$) to 4.19 M ($|V|=8192$).

Quantization-scheme axis. We define five post-training group-quantization schemes applied to all $\text{ndim}=2$ matrices: **int4-g64**, **int6-g128**, **int6-g64**, **int6-g64-mixed** (int6 with substring-matched int8 fallback for embedding/head), and **int8** (requested $g=4096$, falling back to the largest divisor of each tensor’s input dimension when not divisible). For each scheme, real bit-packed payloads are produced and zlib-compressed; reported sizes include code overhead. The full sweep comprises 4 vocabularies $\times$ 5 schemes = 20 cells. Two codec specifications interact non-trivially with the tied-embedding architecture and are diagnosed empirically in Section 4.2.

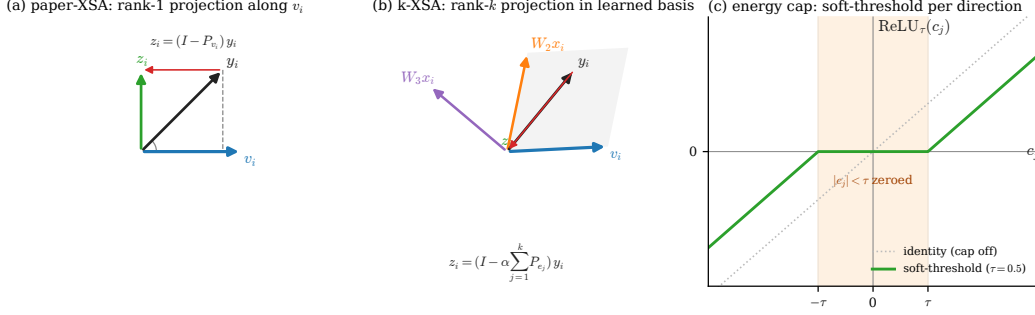


Figure 2: k-XSA geometry. **(a)** Paper-XSA removes the rank-1 component along v_i . **(b)** k-XSA removes a rank- k component anchored at v_i with $W_j x_i$ directions. **(c)** Energy cap zeroes coefficients with $|c_j| < \tau$.

3.4 k-XSA: a rank- k generalization of XSA

3.4.1 Reframing paper-XSA as a rank-1 projection

Apple’s Exclusive Self-Attention Zhai [2026] removes the component of the attention output along v_i as redundant with the FFN. For causal output $y_i = \sum_j a_{ij} v_j$, paper-XSA defines

$$z_i = y_i - \frac{\langle y_i, v_i \rangle}{\|v_i\|^2} v_i = (I - P_{v_i}) y_i, \quad (5)$$

a rank-1 orthogonal projection along v_i (Figure 2a). This implicitly assumes the redundant subspace is one-dimensional and lies along v_i (a direction trained for context aggregation, not FFN overlap) both testable once the projection is generalized.

3.4.2 Definition of k-XSA

Let $B_i \in \mathbb{R}^{k \times d_h}$ collect k candidate basis vectors per head, and $S_i = \text{orth}(B_i)$ its orthonormalization (modified Gram–Schmidt in fp32 to avoid bf16 instability for near-parallel bases). The k-XSA output is

$$z_i = y_i - \alpha_h \sum_{e \in S_i} \langle y_i, e \rangle e = (I - \alpha_h P_{S_i}) y_i, \quad (6)$$

reducing to Equation (5) bit-exactly when $k=1$, $B_i = [v_i]$, $\alpha_h = 1$. Three basis recipes:

$$\begin{aligned} \text{ANCHORED (V+X)} : \quad B_i &= [v_i, W_2 x_i, \dots, W_k x_i] \\ \text{FREE (X)} : \quad B_i &= [W_1 x_i, \dots, W_k x_i] \\ \text{PAPER (V)} : \quad B_i &= [v_i], \quad k = 1. \end{aligned} \quad (7)$$

The $W_j \in \mathbb{R}^{d_h \times D}$ act on residual-stream x_i rather than the GQA value v_i , so their dimensionality is independent of the key/value head count.

3.4.3 Per-head α and the weight-decay fix

Learnable α_h has a confound: AdamW weight decay pulls parameters toward zero, not initialization. Initializing $\alpha_h = 1$ and routing it to the default scalar group (wd = 0.02) biases the optimizer toward $\alpha = 0$, an artefact resembling “learnable α prefers no XSA.” We isolate α_h (and τ below) into a dedicated group with wd = 0 and independent learning rate, applied before any α -related run.

3.4.4 Energy cap: per-direction soft thresholding

Beyond the global α_h , we let the model decide per basis direction how much coefficient to keep:

$$c_j = \langle y_i, e_j \rangle, \quad \tilde{c}_j = \text{sign}(c_j) \cdot \text{ReLU}(|c_j| - \tau_{h,j}), \quad z_i = y_i - \alpha_h \sum_j \tilde{c}_j e_j, \quad (8)$$

Table 1: Ablation summary. All entries: seed 1337, 2000 iters, single H200, post-quant (int8+zlib) val_bpb. Winner (NOSKIP) in bold.

Variant	block params	val_bpb	Δ vs base	submission
Baseline	16.5M	1.29635		14.34 MB
RECURRENT	1.84M	1.51011	+0.21376	2.06 MB
SUBSPACE $k=D/2$	8.87M	1.38957	+0.09322	9.81 MB
AGGREGATE	16.5M	1.29743	+0.00108	14.38 MB
NOSKIP	16.5M	1.29323	-0.00312	14.32 MB

with $\tau_{h,j} \in \mathbb{R}_{\geq 0}$ per-head, per-rank, learnable, zero-initialized, in the same no-WD group as α . Recovers plain k-XSA at $\tau = 0$ and disables projection at $\tau = \infty$ (Figure 2c). Intuition: small coefficients may be “natural” attention–FFN overlap not worth removing; only large overlaps deserve clipping.

We sweep $k \in \{1, 2, 3, 4\}$, the three basis recipes of Equation (7), fixed vs. free α_h , and energy cap on/off; results in Section 4.3.

4 Experiments

We report results in three parts. Section 4.1 evaluates the five architectural variants of Section 3.2. Section 4.2 sweeps the joint design space of tokenizer vocabulary $\times$ post-training quantization scheme. Section 4.3 evaluates k-XSA along its three ingredients rank, basis recipe, and energy cap. All numbers are post-quantization val_bpb on the FineWeb validation set, single-seed unless otherwise noted.

4.1 Architectural ablation

Table 1 summarizes the five variants.

What the ablation rules out. Three findings each falsify a candidate source of headroom. (i) *Iterative refinement is rejected.* RECURRENT loses by +0.214 BPB; the baseline learns a $33\times$ ramp in $\sigma_\ell^{\text{attn}}$ across nine blocks ($0.144 \rightarrow 4.563$), while a shared scale converges to the geometric mean (0.359) and cannot reproduce both endpoints. Depth at this scale is a *specialization axis, not a time axis*. (ii) *Subspace bottlenecks trade BPB for bytes* roughly linearly in parameter savings, never beating baseline at equal cost. (iii) *Learned connectivity does not surface as BPB.* AGGREGATE’s connectivity matrix abandons U-Net cross-skips and converges to “every block reads h_0 plus its own residual” (mean h_0 weight 1.37) real but redundant, since the baseline’s scalar μ_1 already exposes h_0 to every block. Consistently, EMBEDGATE’s per-channel version is BPB-neutral (+0.0009).

What NOSKIP reveals. Removing the U-Net branch shifts the network into a different convergence regime: the attention ramp flattens by $\sim 30\%$ in middle/late layers, the block-8 MLP scale collapses ($2.38 \rightarrow 0.84$), and $|\mu_1|$ in middle blocks grows. The residual path absorbs work previously routed through skips; this less-specialized but better-calibrated regime generalizes marginally better.

4.2 Vocabulary $\times$ quantization sweep

Sweeping the four-vocabulary $\times$ five-scheme grid of Section 3.3 under the held-fixed 9-block baseline, the single legal optimum under the 16 MB cap is **SP2048·int8 at 1.2211 BPB and 15.61 MB**; larger-vocabulary cells with lower pre-cap bpb (SP4096·int8 at 17.11 MB; SP8192·int8 at 19.03 MB) exceed the cap by 7–19% and are inadmissible. Vocabulary dominates quantization by an order of magnitude: at fixed scheme, SP1024 $\rightarrow$ SP8192 drops val_bpb by ~ 0.10 for legal int6/int8, while int8 $\rightarrow$ int4 costs +0.50 to +0.85 BPB at fixed vocabulary, so *the first design choice under the cap is the tokenizer*. int4·g64 is disastrous (1.95–2.10 BPB at $|V| \leq 4096$), only becoming viable at SP8192 where the larger embedding absorbs the coarser grid; by contrast, int8 vs. int6·g64 stays under 0.01 BPB at every vocabulary, with SP8192·int6 landing at 1.22 BPB / 15.18 MB group-wise int6 is essentially free. Two codec patterns surface from the grid without changing the headline: int6·g64-mixed is bit-identical to int6·g64 on tied-embedding models (the (“embed”, “lm_head”) substring match

deeper layers consistently learn larger $\bar{\alpha} \Rightarrow$ model independently rediscovers self-similarity bias

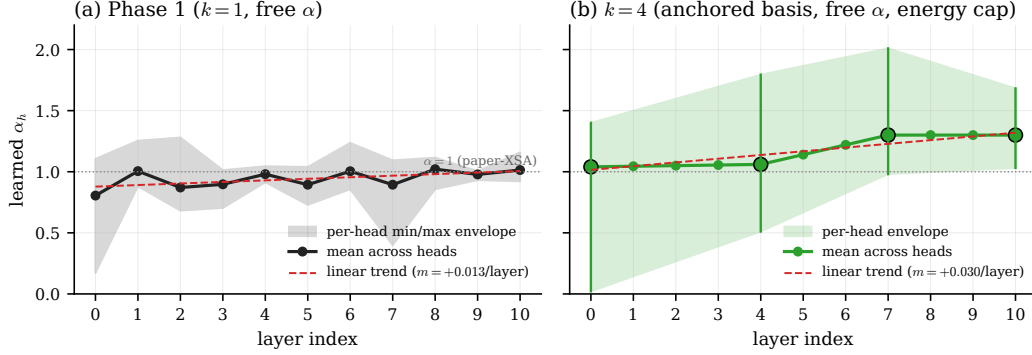


Figure 3: Per-head α across depth. **(a)** $k=1$, free α : wide envelope, $\bar{\alpha}$ slope $+0.013/\text{layer}$, BPB gain over fixed $\alpha=1$ is zero. **(b)** $k=4$: spread widens, slope doubles to $+0.030/\text{layer}$; per-head α now tunes *which* direction to remove.

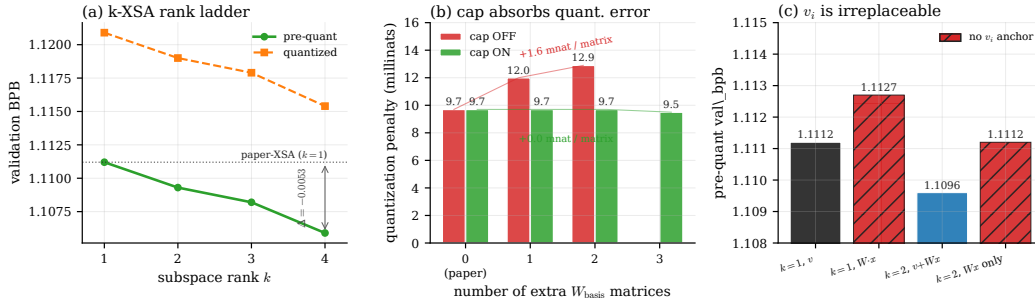


Figure 4: k-XSA rank, quantization, and basis ablation. **(a)** val_bpb decreases monotonically with k (-0.0053 pre-quant, -0.0055 post-quant at $k=4$). **(b)** Energy cap holds the quantization penalty flat at ~ 9.7 mnat regardless of k . **(c)** $W \cdot x$ cannot replace v_i ; the anchored $v+W \cdot x$ basis wins.

fails against tok_emb.weight), and int8 at $g=4096$ silently falls back to $g=128$ via the codec’s largest-divisor rule.

4.3 k-XSA: rank, basis, and energy cap

We evaluate k-XSA along three axes: per-head α at $k=1$, basis recipe at $k \in \{1, 2\}$, and rank ladder $k \in \{1, 2, 3, 4\}$ with the full recipe. Figures 3 and 4 show diagnostics; Table 2 consolidates configurations.

4.3.1 Per-head α at $k=1$ is BPB-neutral; v_i is irreplaceable

With the WD confound (Section 3.4.3) neutralized, free per-head α at $k=1$ yields $\bar{\alpha}_\ell$ growing $+0.013/\text{layer}$ (Figure 3a), corroborating Zhai [2026]’s Figure 1 (deeper layers carry more self-similarity). Yet the BPB delta vs. fixed $\alpha=1$ is zero to three sig figs (1.1112 both): per-head *magnitude* is the wrong knob; per-head *direction* requires $k \geq 2$. At equal added parameters, fully-learned $W \cdot x$ at $k=1$ *loses* $+0.0015$ BPB to paper-XSA (Figure 4c); at $k=2$, two free $W \cdot x$ directions only tie paper, while anchored $v+x$ wins (-0.0016). Thus v_i is necessary but not sufficient consistent with the geometric argument that v_i is trained for context aggregation while the orthogonal complement of the FFN-redundant subspace is not.

4.3.2 Rank ladder and energy-cap quantization absorption

With the full recipe (anchored basis, free α_h , cap on), pre-quant val_bpb improves monotonically from 1.1112 ($k=1$, paper) to 1.1059 ($k=4$): cumulative -0.0053 pre, -0.0055 post-quant (Figure 4a),

Table 2: k-XSA configurations and BPB outcomes. All runs: 3000-step EMA recipe; quantization GPTQ-int6 + brotli; mnat = millinats (post-pre val_bpb).

ID	Config (rank, basis, α , cap)	params	pre-quant	post-quant	quant pen.	artifact
B	$k=1$, v , $\alpha=1$, off (paper)	35.94 M	1.1112	1.1209	+9.7 mnat	16.07 MB
E1	$k=1$, $W \cdot x$, $\alpha=1$, off	37.39 M	1.1127	1.1247	+12.0 mnat	16.68 MB
E3	$k=2$, $W \cdot x$ only, $\alpha=1$, off	38.83 M	1.1112	1.1241	+12.9 mnat	17.30 MB
E5	$k=2$, $v+x$, free α , on	37.39 M	1.1093	1.1190	+9.7 mnat	16.68 MB
k3	$k=3$, $v+x$, free α , on	38.83 M	1.1082	1.1179	+9.7 mnat	17.30 MB
k4	$k=4$, $v+x$, free α , on	40.27 M	1.1059	1.1154	+9.5 mnat	17.92 MB

matching the leaderboard’s 0.005-nat single-seed significance threshold; no saturation by $k=4$. The per-head α heterogeneity that was BPB-neutral at $k=1$ becomes productive at higher rank: spread widens (Figure 3b; block 7 spans [0.98, 2.01]), slope steepens to +0.030/layer, and per-head α tunes *how much* of *which* direction.

The energy cap’s most consequential property was unanticipated: it acts as a *quantization-aware regularizer* (Figure 4b). Without it, each additional W_j adds $\sim +1.6$ mnat to the quant penalty ($+9.7 \rightarrow +12.0 \rightarrow +12.9$ mnat); with it, the penalty stays flat at $\sim +9.7$ mnat for all k . Mechanism: Equation (8) drives small $|c_j|$ to exactly zero, decoupling behavior from the regime where GPTQ degrades W_j most aggressively; free α_h rescales around any quant-induced shift in projector norm.

4.3.3 The combination is what matters

Rank, basis, and per-head magnitude compose to produce the -0.0053 BPB gain; none does so in isolation. Free α at $k=1$ is BPB-neutral; replacing v_i with $W \cdot x$ at $k=1$ is BPB-negative; rank without anchoring is BPB-neutral (Table 2 row E3). Each piece earns its place by enabling the next.

5 Discussion and Conclusion

What the three studies imply for $\mathcal{L}(N)$ at this scale. The three studies decompose the Parameter Golf optimization problem into three semi-independent axes: topology, byte budget, and operator. The architectural ablation (Section 4.1) shows the topology axis is essentially exhausted at this scale, with no rearrangement of the standard transformer skeleton recovering more than 0.003 BPB. The vocabulary-quantization sweep (Section 4.2) identifies the vocabulary as the dominant byte-budget lever, shifting val_bpb by an order of magnitude more than the quantization axis. k-XSA (Section 4.3) shows the operator axis remains productive, yielding a measurable -0.0053 BPB gain on top of any byte-budget operating point. The decomposition matters operationally: without it, comparing across submissions risks attributing operator-axis gains to byte-budget choices and vice versa.

References

- [1] J. Ainslie, J. Lee-Thorp, M. de Jong, Y. Zeiler, S. Sanghai, and Y. Xu. GQA: Training generalized multi-query transformer models from multi-head checkpoints. *arXiv preprint arXiv:2305.13245*, 2023.
- [2] R. Eldan and Y. Li. TinyStories: How small can language models be and still speak coherent English? *arXiv preprint arXiv:2305.07759*, 2023.
- [3] E. Frantar, S. Ashkboos, T. Hoefler, and D. Alistarh. GPTQ: Accurate post-training quantization for generative pre-trained transformers. *arXiv preprint arXiv:2210.17323*, 2022.
- [4] S. Gunasekar, Y. Zhang, J. Aneja, C. C. T. Mendes, A. Del Giorno, S. Gopi, M. Javaheripi, P. Kauffmann, G. de Rosa, O. Saarikivi, et al. Textbooks are all you need. *arXiv preprint arXiv:2306.11644*, 2023.
- [5] J. Hoffmann, S. Borgeaud, A. Mensch, E. Buchatskaya, T. Cai, E. Rutherford, D. de Las Casas, L. A. Hendricks, J. Welbl, A. Clark, et al. Training compute-optimal large language models. *arXiv preprint arXiv:2203.15556*, 2022.

- [6] B. Jacob, S. Kligys, B. Chen, M. Zhu, M. Tang, A. Howard, H. Adam, and D. Kalenichenko. Quantization and training of neural networks for inference at integer arithmetic speed. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 2704–2713, 2018.
- [7] J. Kaplan, S. McCandlish, T. Henighan, T. B. Brown, B. Chess, R. Child, S. Gray, A. Radford, J. Wu, and D. Amodei. Scaling laws for neural language models. *arXiv preprint arXiv:2001.08361*, 2020.
- [8] A. Karpathy. NanoGPT speedrun. <https://github.com/karpathy/nanoGPT>, 2024.
- [9] Z. Liu, C. Zhao, F. Iandola, C. Lai, Y. Tian, I. Fedorov, Y. Xiong, E. Chang, Y. Shi, R. Krishnamoorthi, et al. MobileLLM: Optimizing sub-billion parameter language models for on-device use cases. *arXiv preprint arXiv:2402.14905*, 2024.
- [10] S. Ma, H. Wang, L. Ma, L. Wang, W. Wang, S. Huang, L. Dong, R. Wang, J. Xue, and F. Wei. The era of 1-bit LLMs: All large language models are in 1.58 bits. *arXiv preprint arXiv:2402.17764*, 2024.
- [11] OpenAI. Parameter golf. <https://github.com/openai/parameter-golf>, 2026.
- [12] G. Penedo, H. Kydliček, A. Cappelli, M. Sasko, and T. Wolf. The FineWeb datasets: Decanting the web for the finest text data at scale. *arXiv preprint arXiv:2406.17557*, 2024.
- [13] N. Shazeer. Fast transformer decoding: One write-head is all you need. *arXiv preprint arXiv:1911.02150*, 2019.
- [14] J. Su, Y. Lu, S. Pan, A. Murtadha, B. Wen, and Y. Liu. RoFormer: Enhanced transformer with rotary position embedding. *arXiv preprint arXiv:2104.09864*, 2022.
- [15] Y. Sun, X. Li, F. Dalvi, J. Koo, G. Bhatt, T. Sheils, A. Gu, and C. Ré. Learning to (Learn at Test Time): RNNs with expressive hidden states. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, 2024.
- [16] H. Wang, S. Ma, L. Dong, S. Huang, D. Wang, and F. Wei. BitNet: Scaling 1-bit transformers for large language models. *arXiv preprint arXiv:2310.11453*, 2023.
- [17] S. Zhai. Exclusive self attention. *arXiv preprint arXiv:2603.09078*, 2026.